Final Integrated Report: WoSt Optimization and Zombie Baseline Comparison

Generated on 2026-06-02.

This final report integrates three sources:

- `C:\THU\homework\zombie\results_zombie_final_report\ZOMBIE_VS_WOST_FINAL_REPORT.md`
- `experiments/optimization_report.md`
- `experiments/formal_comparison_report.md`

When older and fresh numbers differ slightly, this report uses the fresh formal runs in `experiments/formal_comparison_report.md` as the primary quantitative reference. The Zombie final report is used as supporting context and confirms the same broad conclusion: Zombie and WoSt agree closely on the clean Dirichlet benchmark, while the mixed Neumann benchmark exposes larger implementation-level differences.

1. Benchmark Setup

All main cross-project comparisons use the Bunny mesh:

```
mesh: Stanford Bunny
vertices: 35,292
faces: 70,580
outer cube half extent: 0.22
domain: outer cube minus inner Bunny mesh
analytic solution: u(x,y,z) = x + y + z
PDE: Delta u = 0
```

Two boundary-condition settings are tested:

```
Dirichlet-only:
  inner Bunny boundary: Dirichlet
  outer cube boundary: Dirichlet
  g(p) = p.x + p.y + p.z

Mixed Neumann:
  inner Bunny boundary: Neumann
  outer cube boundary: Dirichlet
  g_outer(p) = p.x + p.y + p.z
  h_inner(p) = grad(u) dot n = (1,1,1) dot n
```

Fresh formal output folders:

```
experiments/formal_wost_dirichlet/results/
experiments/formal_wost_neumann/results/
experiments/formal_zombie_dirichlet/
experiments/formal_zombie_neumann/
```

WoSt-only optimization outputs:

```
experiments/optimization_summary.csv
experiments/optimization_points.csv
experiments/optimization_report.md
experiments/figures/
```

2. Main Conclusions

1. On the clean Dirichlet-only Bunny benchmark, WoSt and Zombie are very close in accuracy. WoSt is marginally more accurate in the fresh formal run, while Zombie is faster through its current Python baseline solve loop.
2. Both solvers show the expected Monte Carlo convergence trend as random walks per point increase.
3. On the mixed Neumann Bunny benchmark, WoSt becomes faster than Zombie and is competitive or better at practical epsilon values such as `1e-4` and `1e-5`.
4. Zombie is more stable at very coarse Neumann epsilon values, especially `epsilon=1e-2`, while WoSt shows large boundary bias at that coarse tolerance.
5. WoSt now includes additional optimization features and diagnostics: relative-standard-error adaptive sampling, antithetic sampling, lazy star-radius refinement statistics, Common Random Numbers, epsilon extrapolation, and a Neumann normal-convention sanity test.
6. The old absolute-standard-error adaptive criterion does not significantly reduce samples on the formal Bunny Dirichlet setup. The new relative-standard-error adaptive criterion reduces samples in the smoke optimization experiment, but at the tested aggressive setting it increases error; it should be tuned for full-scale production experiments.

Recommended final wording:

```
We compared our WoSt implementation against Zombie on a 70,580-face Bunny mesh
using the analytic solution u=x+y+z. On the Dirichlet-only benchmark, both
solvers closely agree and show the expected Monte Carlo convergence trend. On the
mixed Neumann benchmark, both solvers converge as walk count increases, but their
reflected-path behavior differs substantially. WoSt uses much shorter average
paths and is faster at practical epsilon values, while Zombie is more stable at
very coarse epsilon. We also added WoSt-only optimization diagnostics for
adaptive sampling, antithetic sampling, lazy star-radius refinement, Common
Random Numbers, epsilon extrapolation, and Neumann normal-convention validation.
```

2.1 Figure Sources

All figures used in this integrated report were copied into one local folder:

```
experiments/final_report_figures/
```

This folder includes:

- fresh formal comparison figures from `experiments/formal_zombie_dirichlet/` and `experiments/formal_zombie_neumann/`
- WoSt optimization figures from `experiments/figures/`
- historical reference figures mentioned in `ZOMBIE_VS_WOST_FINAL_REPORT.md`

3. Formal Dirichlet Comparison

Formal setup:

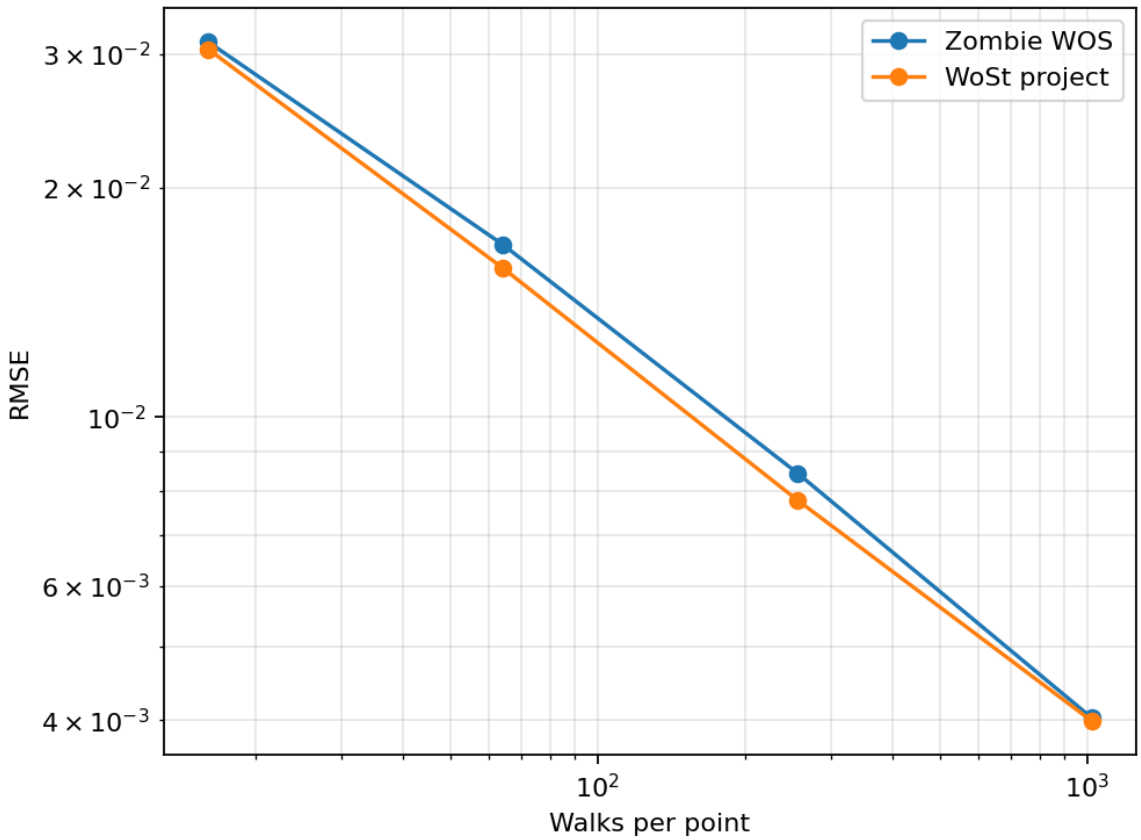
```
queries = 500
grid = 16^3
seed = 12345
cube half extent = 0.22
epsilon for convergence = 1e-4
max_steps = 512
```

3.1 Convergence

Walks	Zombie RMSE	WoSt RMSE	Zombie / WoSt RMSE	Zombie time (s)	WoSt time (s)
16	0.03110	0.03042	1.022	0.24	0.85
64	0.01685	0.01570	1.073	0.99	3.31
256	0.00845	0.00778	1.085	3.88	13.18
1024	0.00403	0.00399	1.010	15.58	53.03

Observation:

- Both methods show clean Monte Carlo convergence.
- WoSt is slightly more accurate at all four sample counts in the fresh formal run.
- Zombie is faster on this Dirichlet solve-loop benchmark.



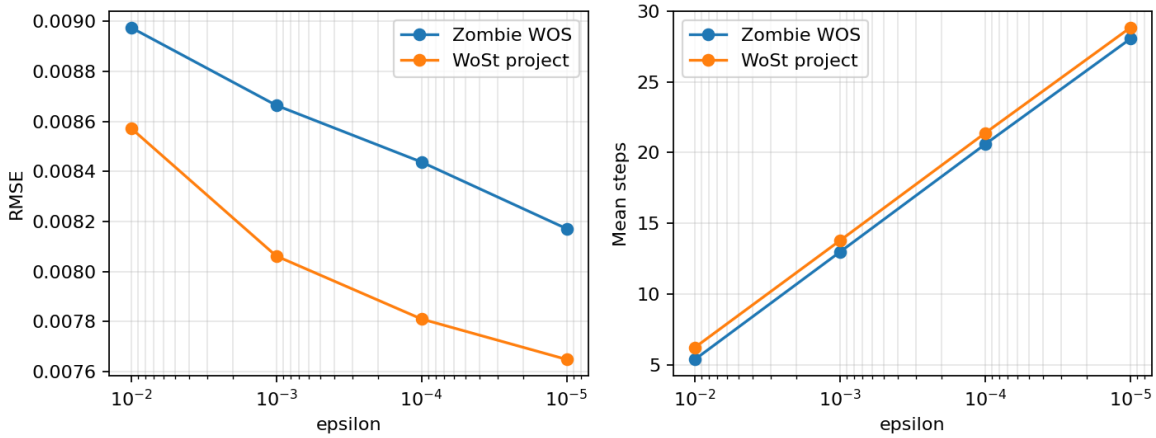
3.2 Epsilon Sweep

Both solvers use `walks_per_point=256`.

Epsilon	Zombie RMSE	WoSt RMSE	Zombie / WoSt RMSE	Zombie time (s)	WoSt time (s)
1e-2	0.00897	0.00857	1.047	1.78	12.35
1e-3	0.00866	0.00806	1.075	3.01	12.90
1e-4	0.00844	0.00781	1.080	3.64	13.27
1e-5	0.00817	0.00765	1.068	4.25	13.62

Observation:

- WoSt has slightly lower RMSE for all tested epsilon values.
- RMSE changes mildly because this linear Dirichlet problem is mainly Monte Carlo noise limited at 256 walks.
- Runtime increases as epsilon tightens, as expected.



3.3 Structured Grid

Metric	Zombie	WoSt
Grid	16^3	16^3
Valid points	4018	4018
RMSE	0.00747	0.00730
Runtime (s)	23.74	87.60

Observation:

- Structured-grid accuracy is effectively aligned.
- WoSt has slightly smaller RMSE.
- Zombie is faster on the Dirichlet grid solve.

VTK outputs:

```
experiments/formal_wost_dirichlet/results/linear_dirichlet_grid.vtk
experiments/formal_zombie_dirichlet/linear_dirichlet_grid.vtk
```

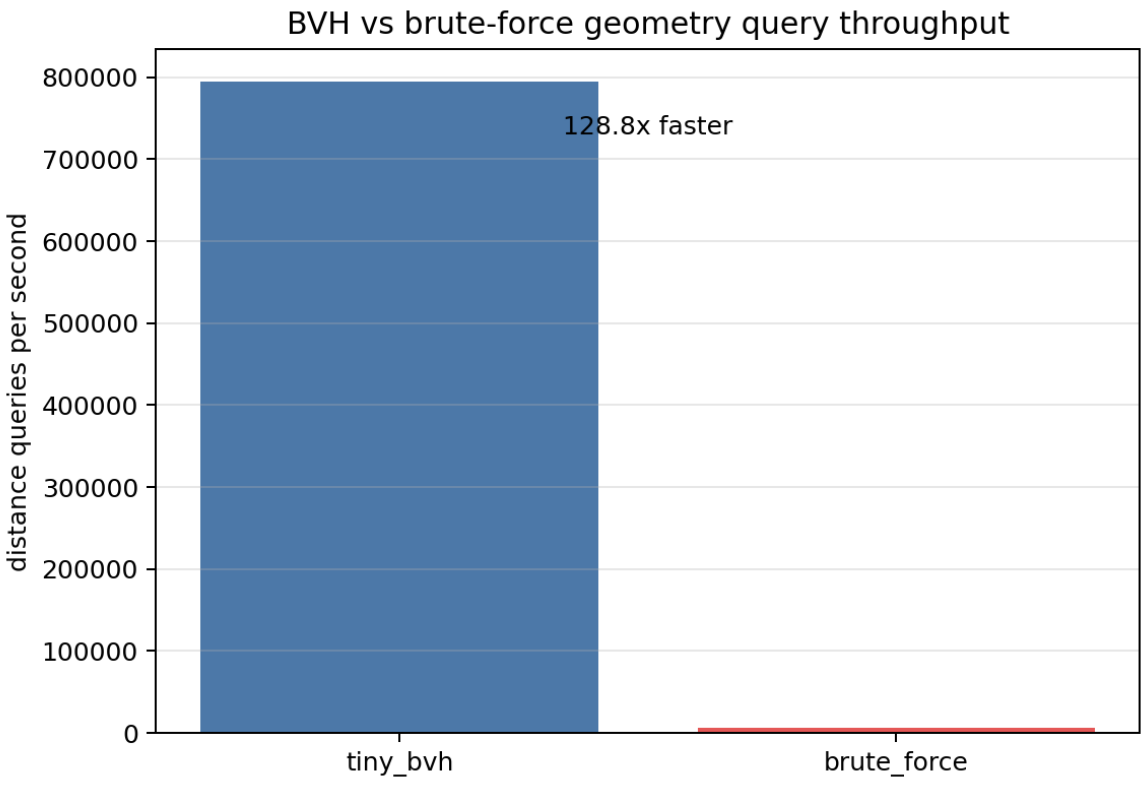
4. Geometry Query Microbenchmark

Backend	Queries	Query/s	Time (s)
WoSt tiny_bvh	500	338,364	0.00148
WoSt brute force	500	5,417	0.09231
Zombie FCPW BVH	500	265,788	0.00188

Observation:

- WoSt `tiny_bvh` is about `62.5x` faster than the WoSt brute-force distance baseline.
- In this exposed distance-query microbenchmark, WoSt `tiny_bvh` is about `1.27x` faster than Zombie FCPW BVH.
- This should not be over-interpreted as a pure library comparison because Zombie is measured through Python-loop calls and the scripts use different geometry-query conventions.

Historical WoSt geometry reference from the Zombie final report:



5. Formal Mixed Neumann Comparison

Formal setup:

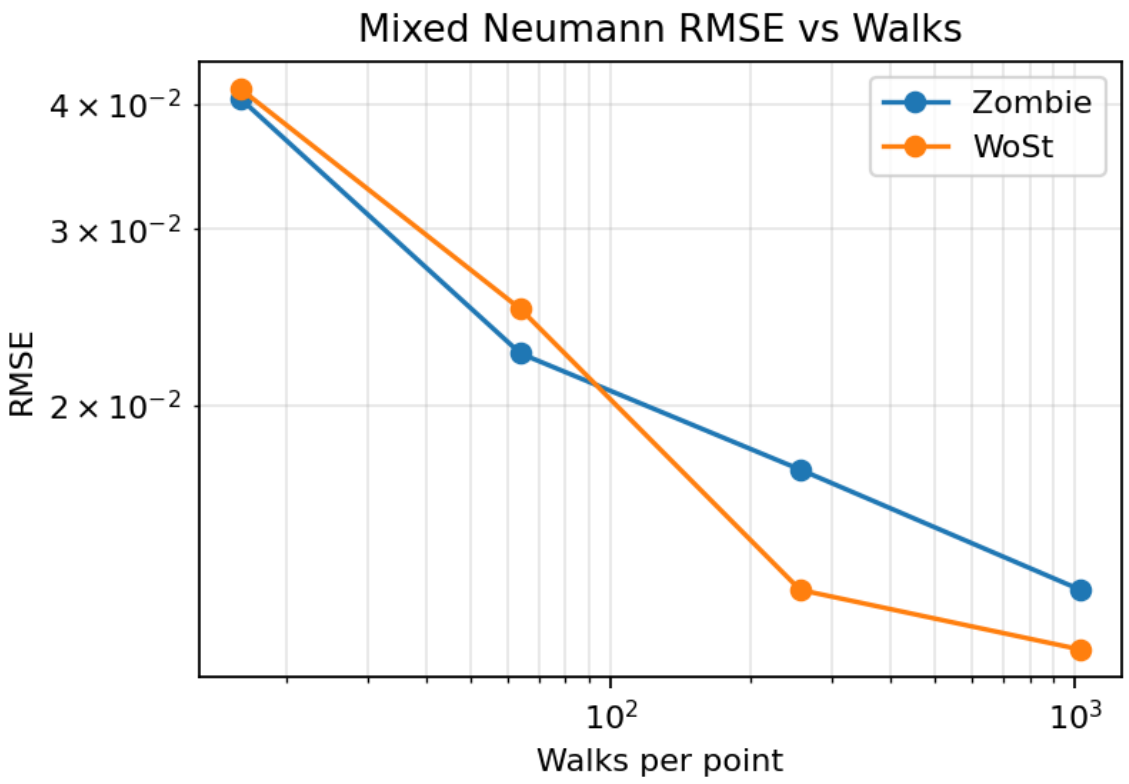
```
queries = 100
grid = 8^3
seed = 32345
cube half extent = 0.22
max_steps = 2048
```

5.1 Convergence

Walks	Zombie RMSE	WoSt RMSE	Zombie / WoSt RMSE	Zombie mean steps	WoSt mean steps	Zombie time (s)	WoSt time (s)
16	0.04048	0.04140	0.978	282.19	30.93	2.06	1.11
64	0.02254	0.02497	0.903	227.22	33.52	6.76	4.44
256	0.01726	0.01308	1.319	238.82	34.46	28.00	17.20
1024	0.01309	0.01141	1.148	239.65	34.47	111.35	71.06

Observation:

- Both methods converge as walk count increases.
- Zombie is slightly better at low walk counts.
- WoSt is better at 256 and 1024 walks.
- Zombie uses much longer average paths, while WoSt's reflected paths are much shorter in this benchmark.



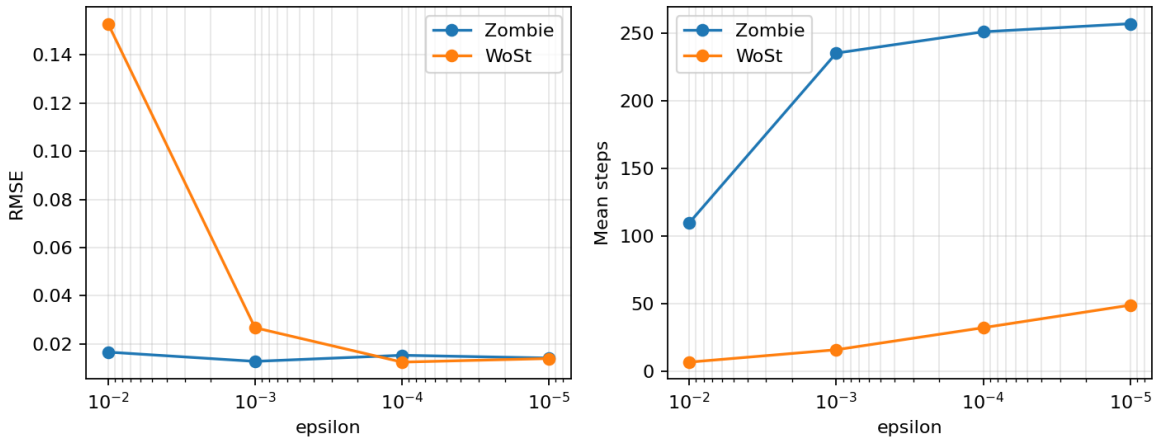
5.2 Epsilon Sweep

Both solvers use `walks_per_point=256`.

Epsilon	Zombie RMSE	WoSt RMSE	Zombie / WoSt RMSE	Zombie mean steps	WoSt mean steps	Zombie time (s)	WoSt time (s)
1e-2	0.01664	0.15294	0.109	109.81	6.75	59.52	9.67
1e-3	0.01280	0.02676	0.478	235.35	15.90	65.15	10.61
1e-4	0.01532	0.01249	1.227	251.10	32.21	29.13	15.15
1e-5	0.01418	0.01398	1.014	257.10	48.88	24.93	17.92

Observation:

- Zombie is much more stable at coarse epsilon values.
- WoSt has severe boundary bias at `epsilon=1e-2`, improves sharply by `1e-4`, and is competitive at `1e-5`.
- WoSt is faster at all four epsilon values in this fresh formal Neumann run.
- This supports the epsilon-extrapolation conclusion: large changes between `epsilon` and `epsilon/2` are a useful signal of boundary bias.



5.3 Structured Grid

Metric	Zombie	WoSt
Grid	8^3	8^3
Valid points	508	508
RMSE	0.01227	0.01196
Mean steps	139.35	19.62
Runtime (s)	82.36	16.14

Observation:

- Neumann grid RMSE is nearly identical.
- WoSt is about `5.1x` faster on this grid run.

- The speed difference is mainly explained by the much smaller WoSt mean step count.

VTK outputs:

```
experiments/formal_wost_neumann/results/neumann_mixed_grid.vtk
experiments/formal_zombie_neumann/neumann_mixed_grid.vtk
```

6. WoSt Optimization Experiments

The WoSt optimization report uses append-only results from:

```
experiments/optimization_summary.csv
experiments/optimization_points.csv
```

The plotting/report script summarizes the latest three rows for each experiment/method pair. These optimization runs are WoSt-only diagnostics, not direct Zombie comparisons.

6.1 Common Random Numbers

Each comparison uses Common Random Numbers:

```
experiment seed -> query points
SeedFor(experiment_seed, point_index, stream) -> per-query random walk stream
```

This reduces comparison noise for:

- fixed vs adaptive sampling
- normal vs antithetic sampling
- full star-radius vs lazy refinement
- epsilon vs epsilon/2

6.2 Adaptive Sampling

New relative-standard-error criterion:

```
relative_standard_error = standard_error / max(abs(mean_estimate), rse_eps)
stop when relative_standard_error < target_rse
```

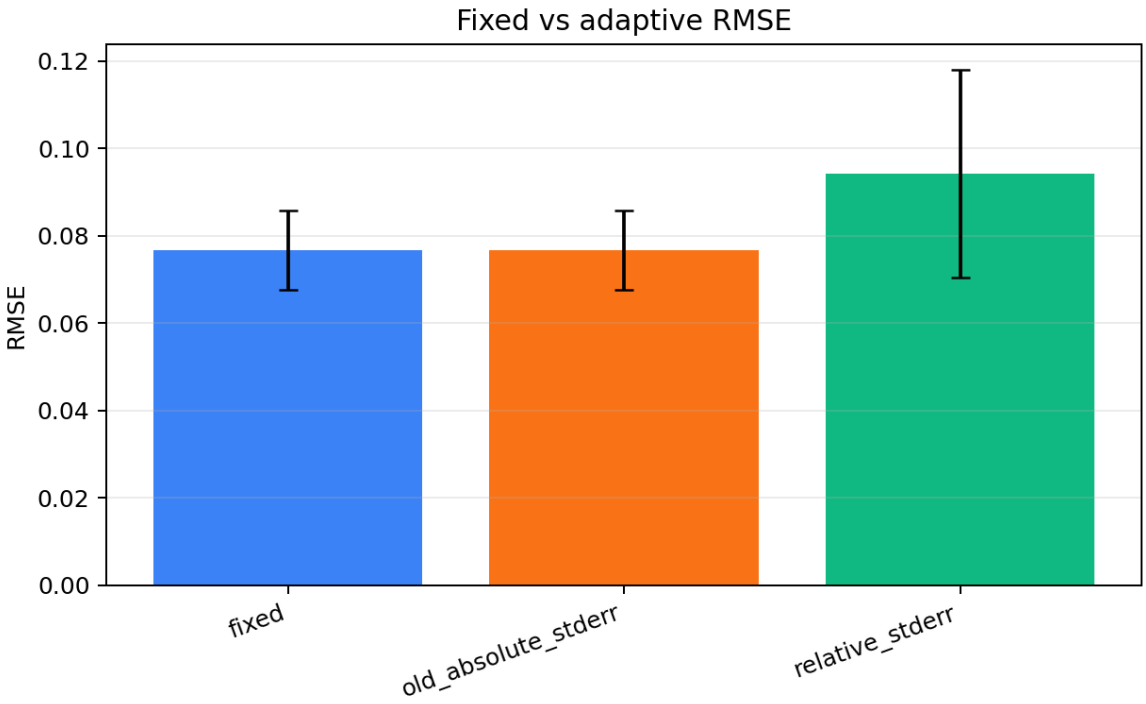
Latest small-scale optimization diagnostic:

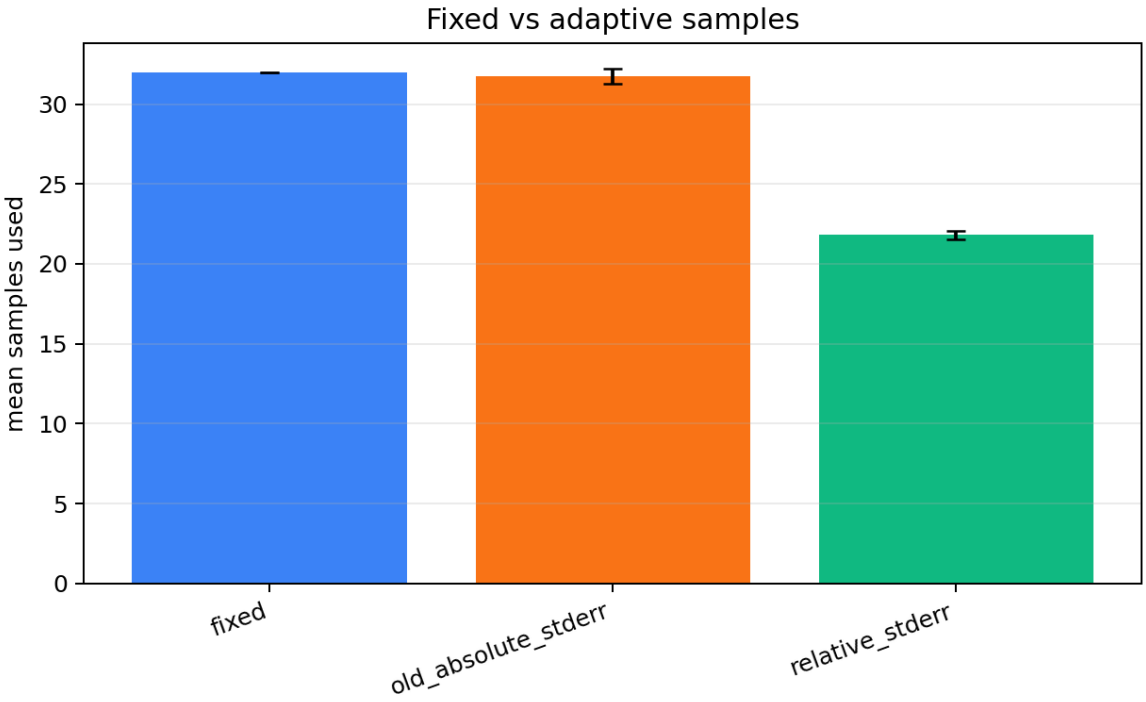
Method	RMSE	MAE	Mean samples	Runtime (s)
fixed	0.07671 +/- 0.00907	0.05505 +/- 0.00725	32.00 +/- 0.00	0.07169 +/- 0.00804
old absolute stderr	0.07671 +/- 0.00907	0.05505 +/- 0.00725	31.73 +/- 0.46	0.07586 +/- 0.01330
relative stderr	0.09416 +/- 0.02370	0.07127 +/- 0.01700	21.81 +/- 0.28	0.05328 +/- 0.00555

Interpretation:

- Relative-standard-error adaptive sampling reduced mean samples by about `31.8%` in the small smoke-scale test.
- The tested threshold was aggressive, so RMSE increased.
- This confirms the new mechanism is effective at reducing sampling, but full-scale tuning is needed to hit the target of lower samples without significant RMSE degradation.
- On the formal Bunny Dirichlet point run, the old absolute criterion reduced samples only from `1024.0` to `997.7`, confirming why the criterion needed to be changed.

Adaptive RMSE and sample-count figures:





6.3 Antithetic Sampling

Antithetic sampling uses paired directions:

```
d and -d
```

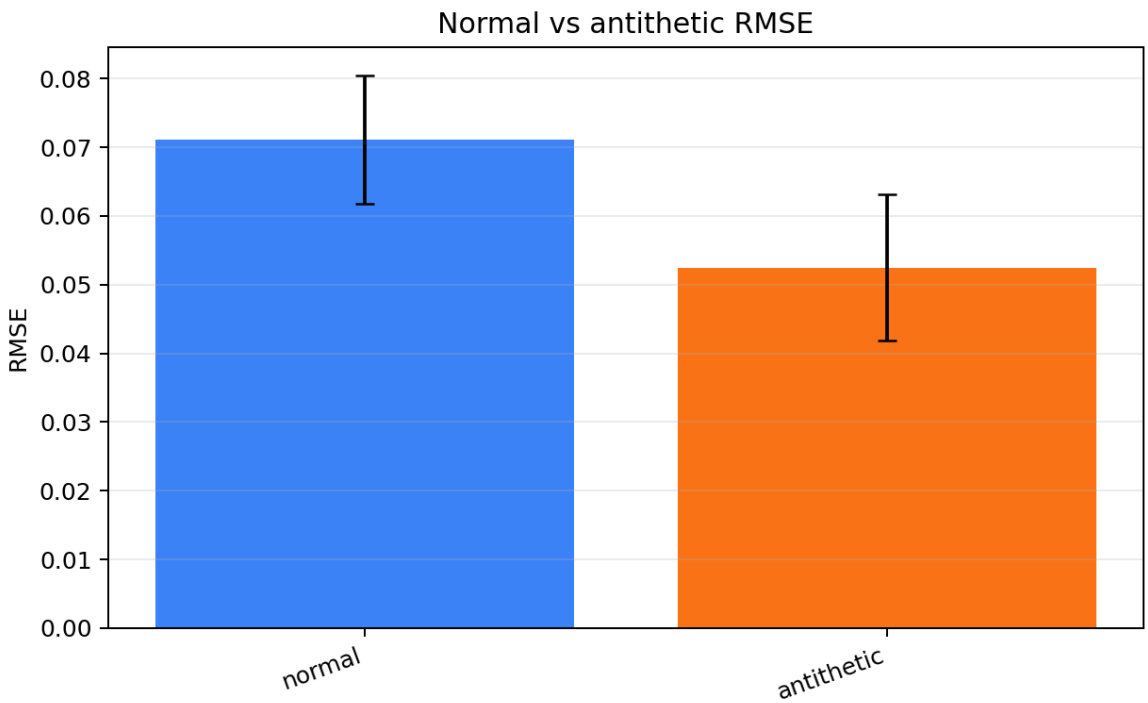
Latest diagnostic:

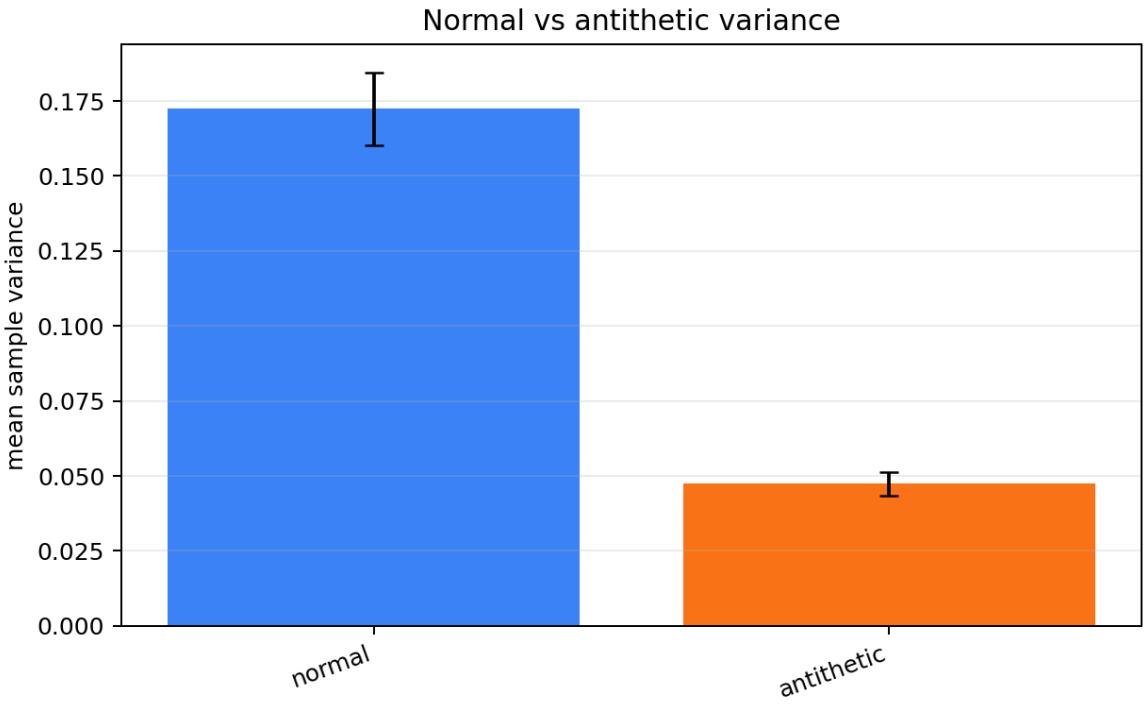
Method	RMSE	MAE	Mean sample variance	Runtime (s)
normal	0.07118 +/- 0.00934	0.05520 +/- 0.00517	see figure	0.07339 +/- 0.00807
antithetic	0.05249 +/- 0.01070	0.03969 +/- 0.00844	see figure	0.07071 +/- 0.00431

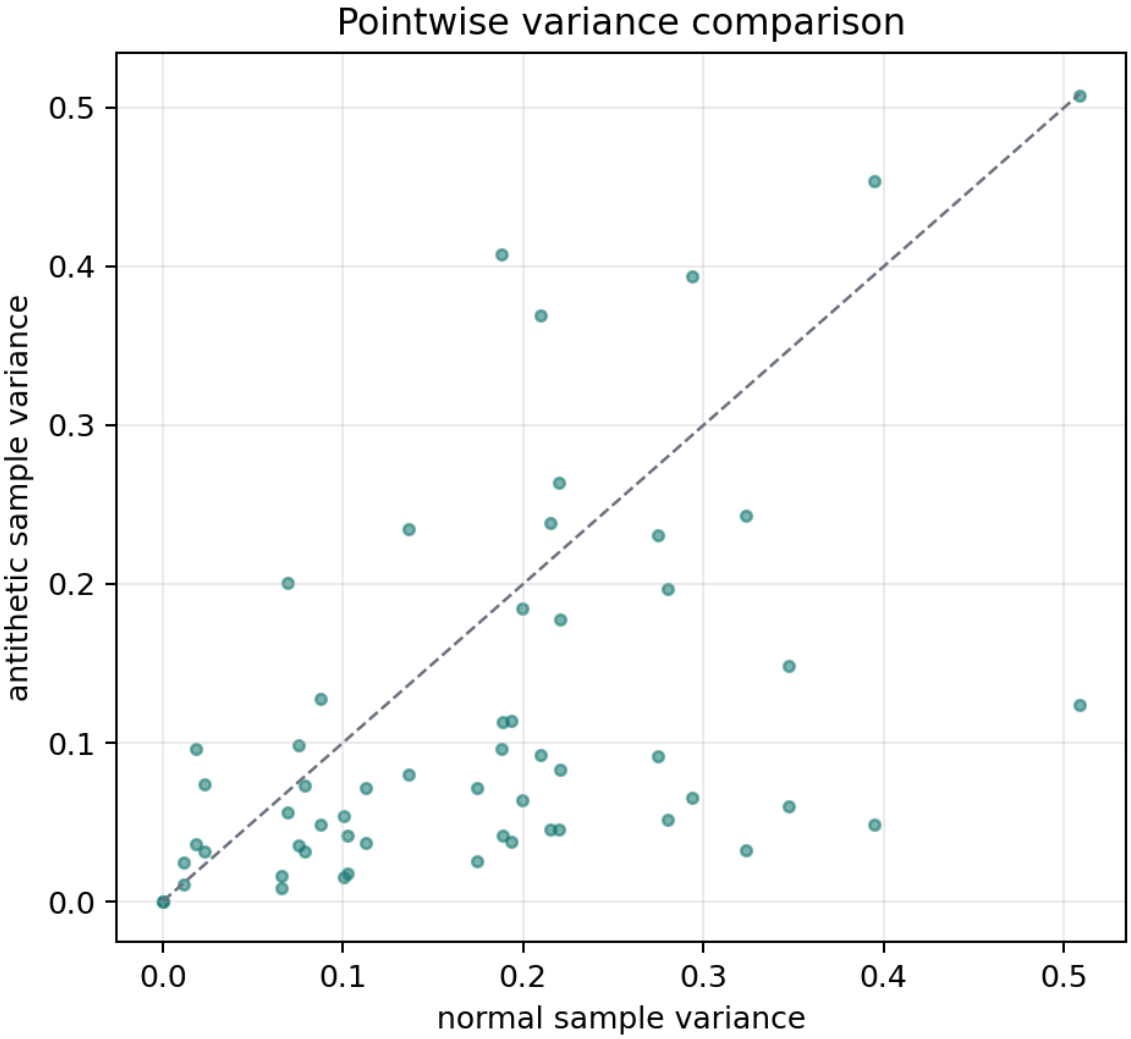
Interpretation:

- Antithetic sampling reduced RMSE and MAE in the small-scale test.
- The paired-estimator variance is lower in the diagnostic plot.
- Runtime stayed comparable.

Antithetic sampling figures:




Normal vs antithetic variance
mean sample variance
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
normal
antithetic


Pointwise variance comparison
antithetic sample variance
normal sample variance
0.0
0.1
0.2
0.3
0.4
0.5

6.4 Lazy Star-Radius Refinement

Lazy refinement compares full exact star-radius queries with thresholded exact refinement.

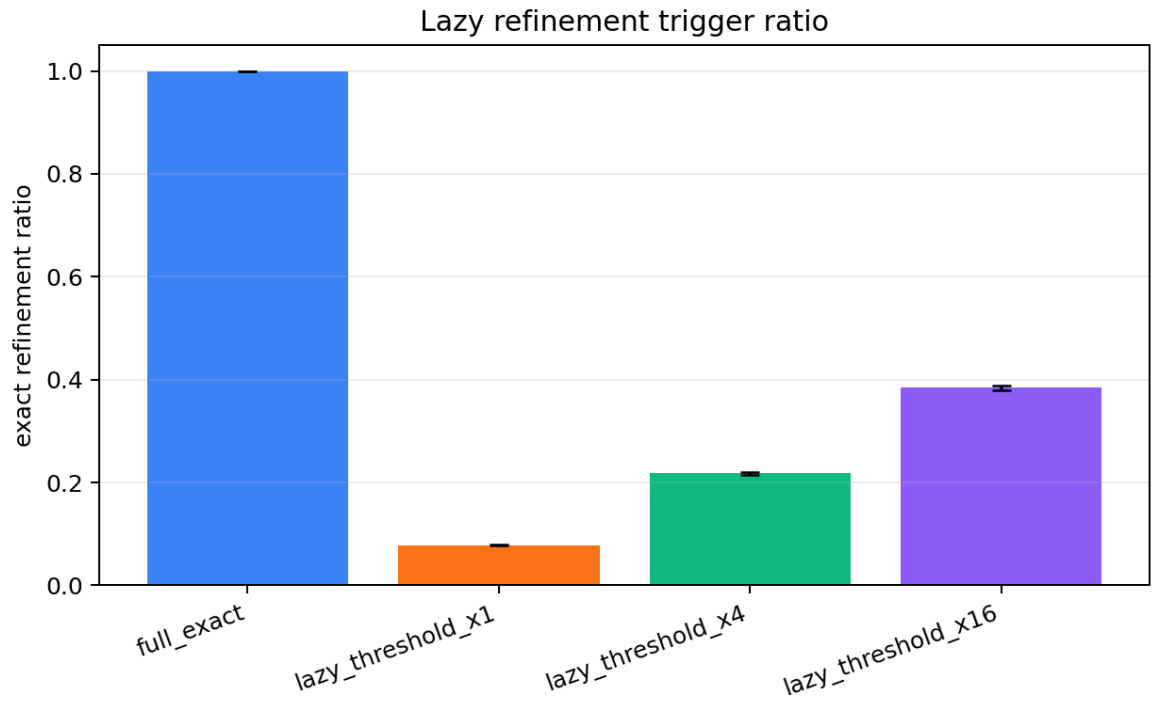
Latest diagnostic:

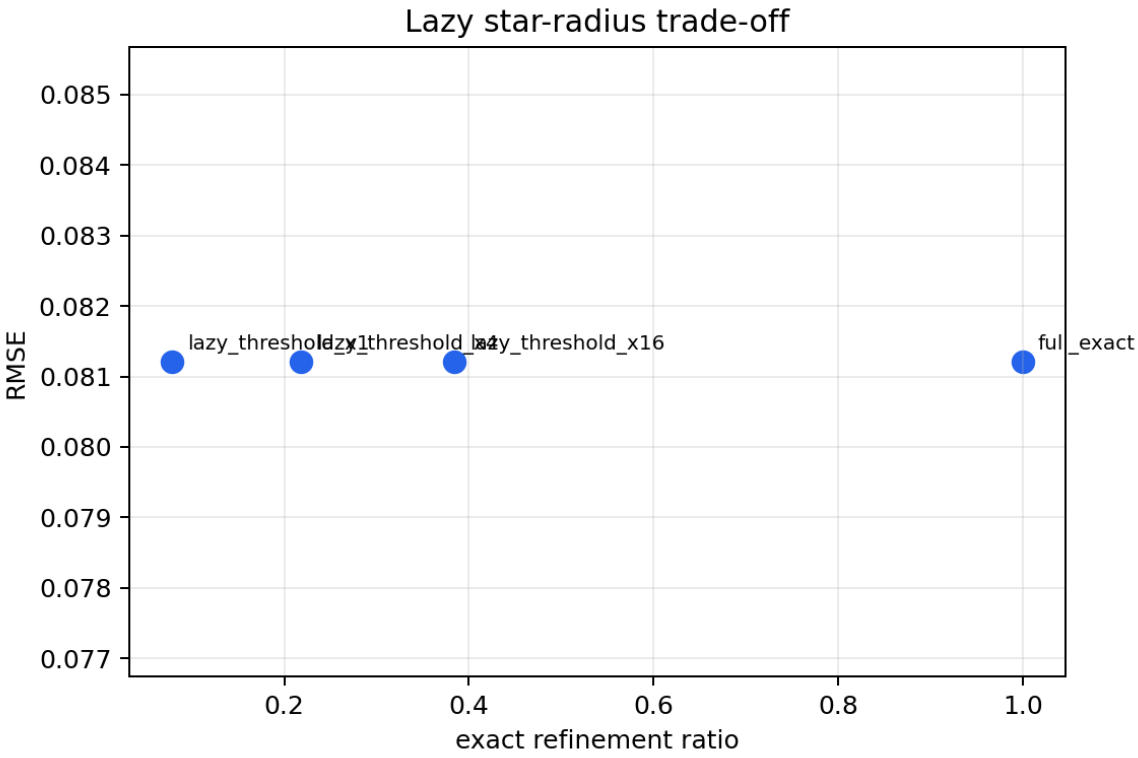
Method	RMSE	MAE	Runtime (s)
full exact	0.08121 +/- 0.01370	0.05682 +/- 0.00695	0.56973 +/- 0.03660
lazy threshold x1	0.08121 +/- 0.01370	0.05682 +/- 0.00695	0.06921 +/- 0.00390
lazy threshold x4	0.08121 +/- 0.01370	0.05682 +/- 0.00695	0.14187 +/- 0.00892
lazy threshold x16	0.08121 +/- 0.01370	0.05682 +/- 0.00695	0.23912 +/- 0.03270

Interpretation:

- Lazy refinement preserved RMSE in the diagnostic run.
- It strongly reduced runtime compared with full exact star-radius.
- Trigger ratios and spatial trigger locations are visualized.

Lazy refinement figures:




Lazy star-radius trade-off
RMSE
exact refinement ratio
0.085
0.084
0.083
0.082
0.081
0.080
0.079
0.078
0.077
0.2
0.4
0.6
0.8
1.0
lazy_threshold_x1
lazy_threshold_x4
lazy_threshold_x16
full_exact


Where lazy refinement triggers
y
x
point refinement ratio
1.0
0.5
0.0
−0.5
−1.0
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
0.11
0.10
0.09
0.08
0.07
0.06

6.5 Epsilon Extrapolation

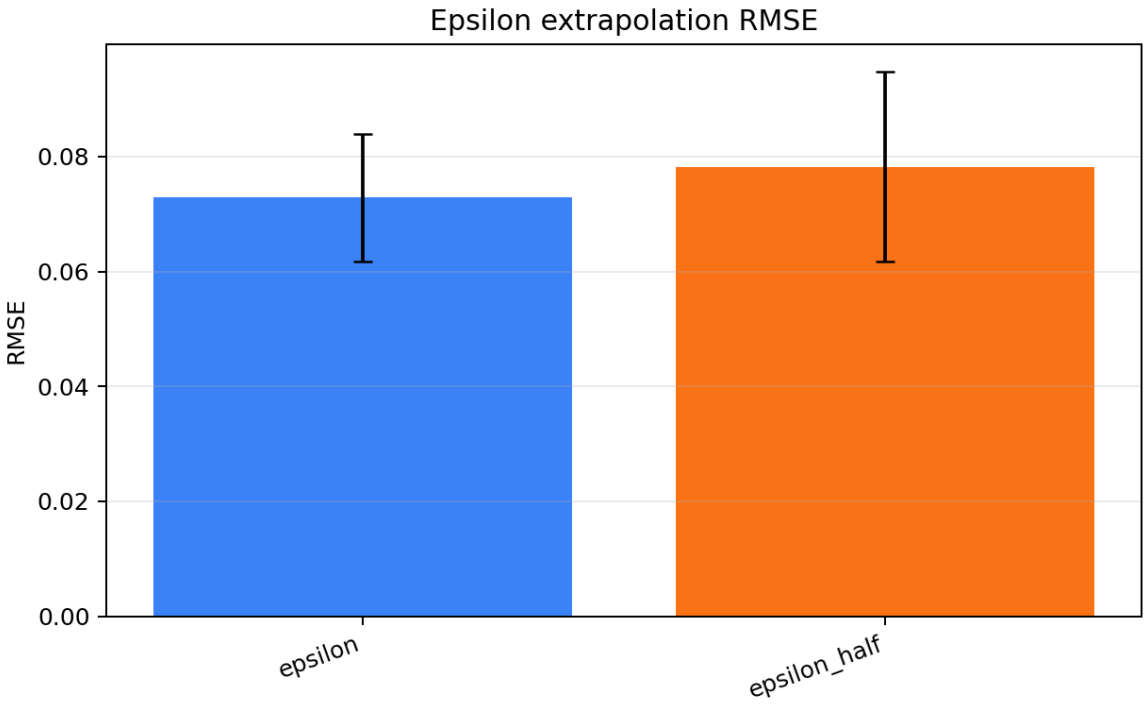
Diagnostic comparison:

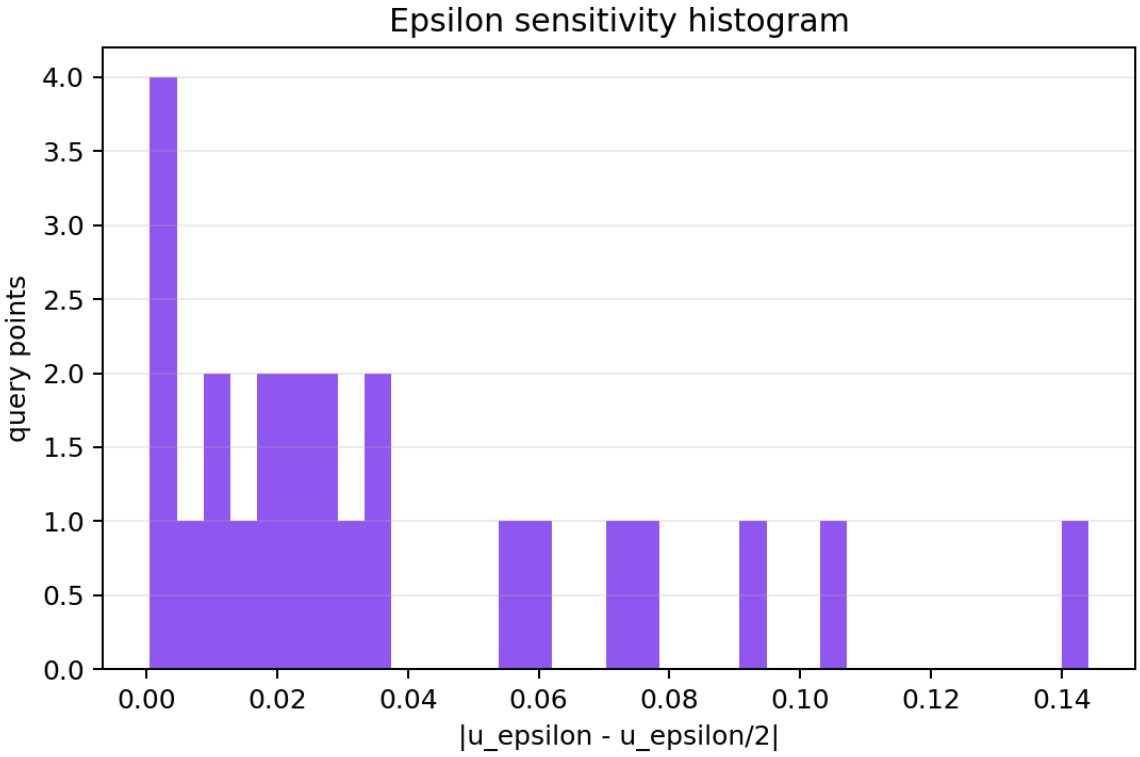
Method	RMSE	MAE	Runtime (s)
epsilon	0.07287 +/- 0.01110	0.05592 +/- 0.00958	0.05560 +/- 0.00414
epsilon_half	0.07824 +/- 0.01650	0.06062 +/- 0.00911	0.05460 +/- 0.00486

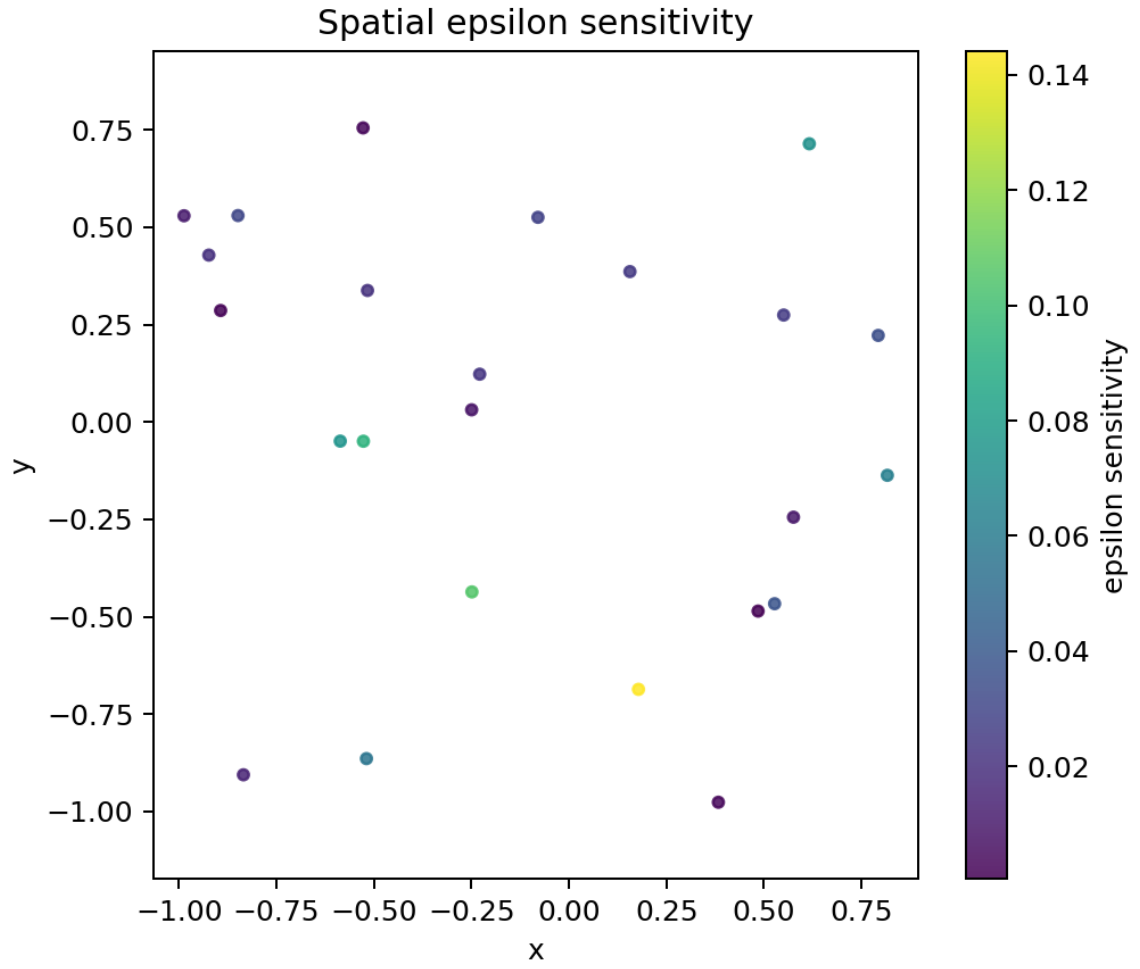
Interpretation:

- Pointwise `|u_epsilon - u_epsilon/2|` is useful for locating epsilon-sensitive regions.
- This supports possible adaptive epsilon refinement near complex Neumann boundaries.
- The formal Neumann epsilon sweep reinforces this: WoSt has high bias at `1e-2` but improves sharply by `1e-4`.

Epsilon extrapolation figures:




Epsilon sensitivity histogram
query points
|u_epsilon - u_epsilon/2|
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14


Spatial epsilon sensitivity
y
x
epsilon sensitivity
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
0.14
0.12
0.10
0.08
0.06
0.04
0.02

6.6 Neumann Normal Convention Sanity Test

The sphere/cube sanity test checks:

```
inner sphere: Neumann boundary
outer cube: Dirichlet boundary
exact solution: u=x+y+z
h = (1,1,1) dot n
```

Latest diagnostic:

Method	RMSE	MAE	Runtime (s)
sphere/cube sanity	0.15580 +/- 0.02700	0.11117 +/- 0.02440	0.06395 +/- 0.01300

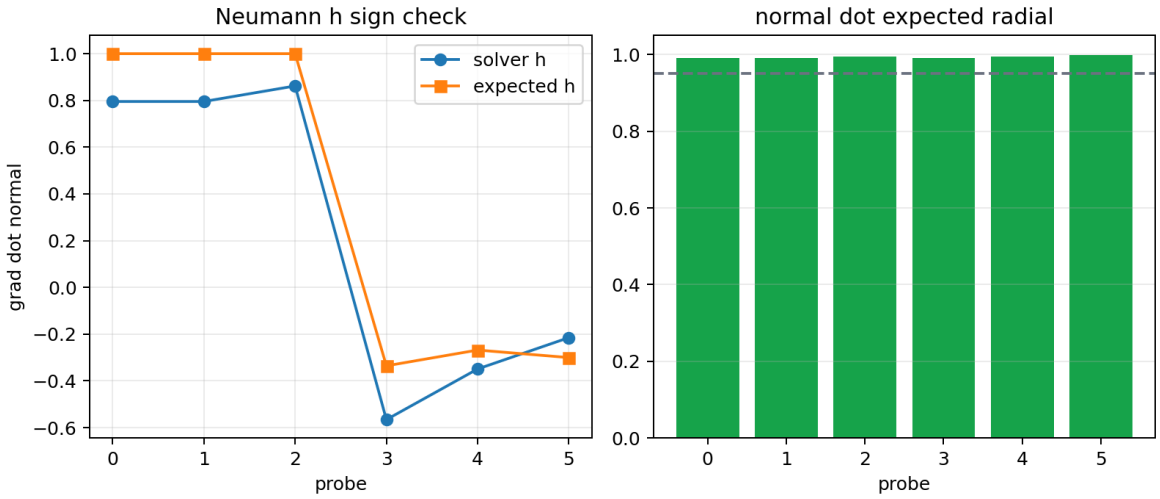
Normal diagnostic:

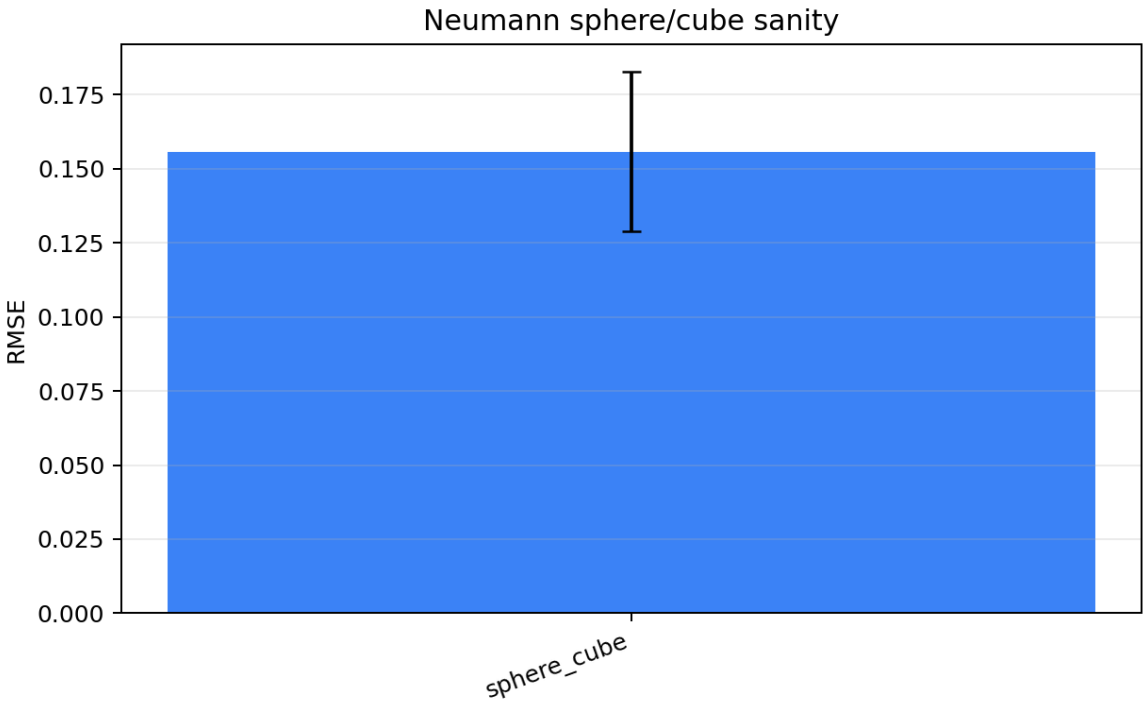
```
mean normal dot expected radial ~= 0.993
min normal dot expected radial  ~= 0.990
status: PASS
```

Interpretation:

- The mesh normal direction and Neumann sign convention are consistent on the generated sphere.
- This supports the claim that the Neumann normal convention was validated before applying the solver to complex Bunny geometry.

Neumann normal convention figures:

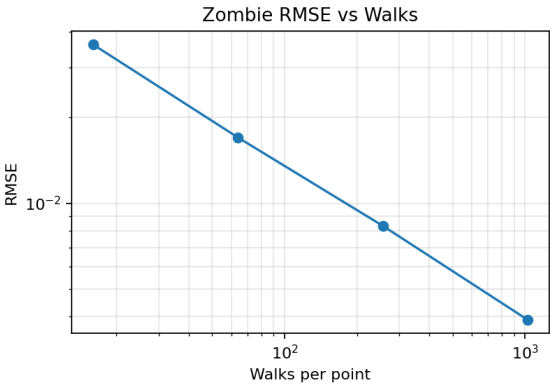
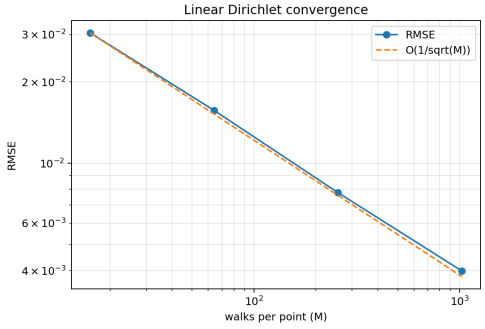
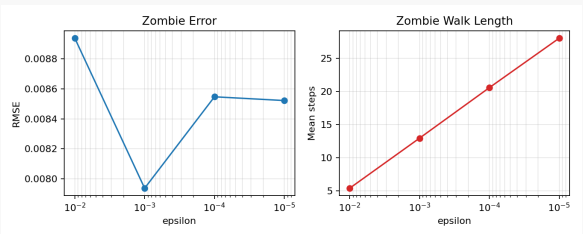
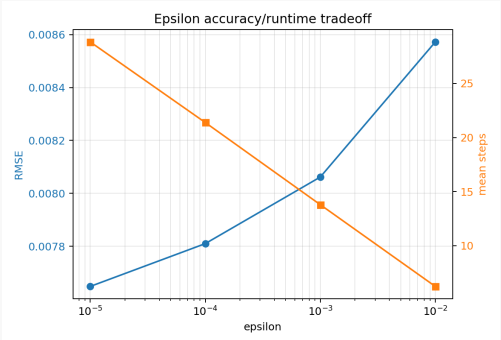




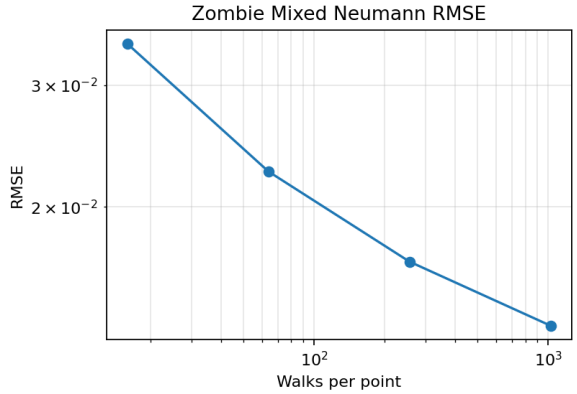
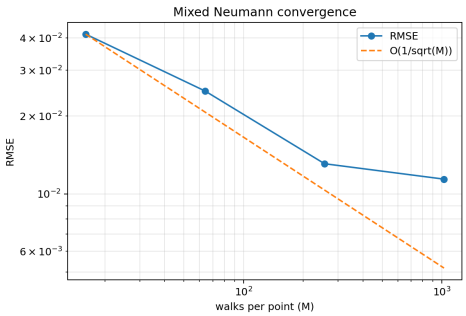
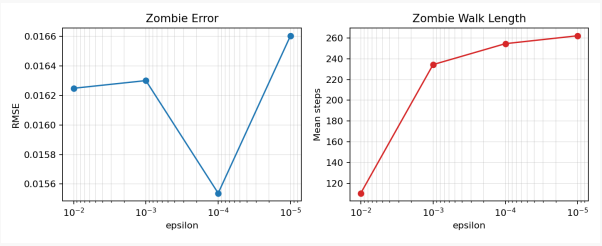
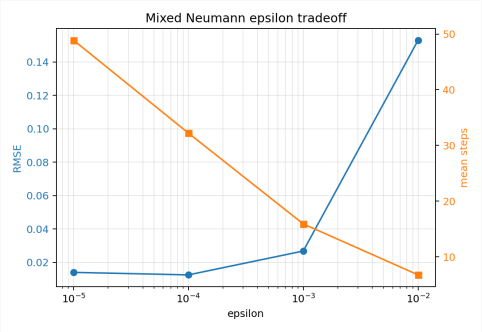
7. Historical Reference Figures from Zombie Final Report

The fresh formal comparison above is the primary quantitative source. The following figures are included because they were referenced in the earlier Zombie final report and are useful visual context.

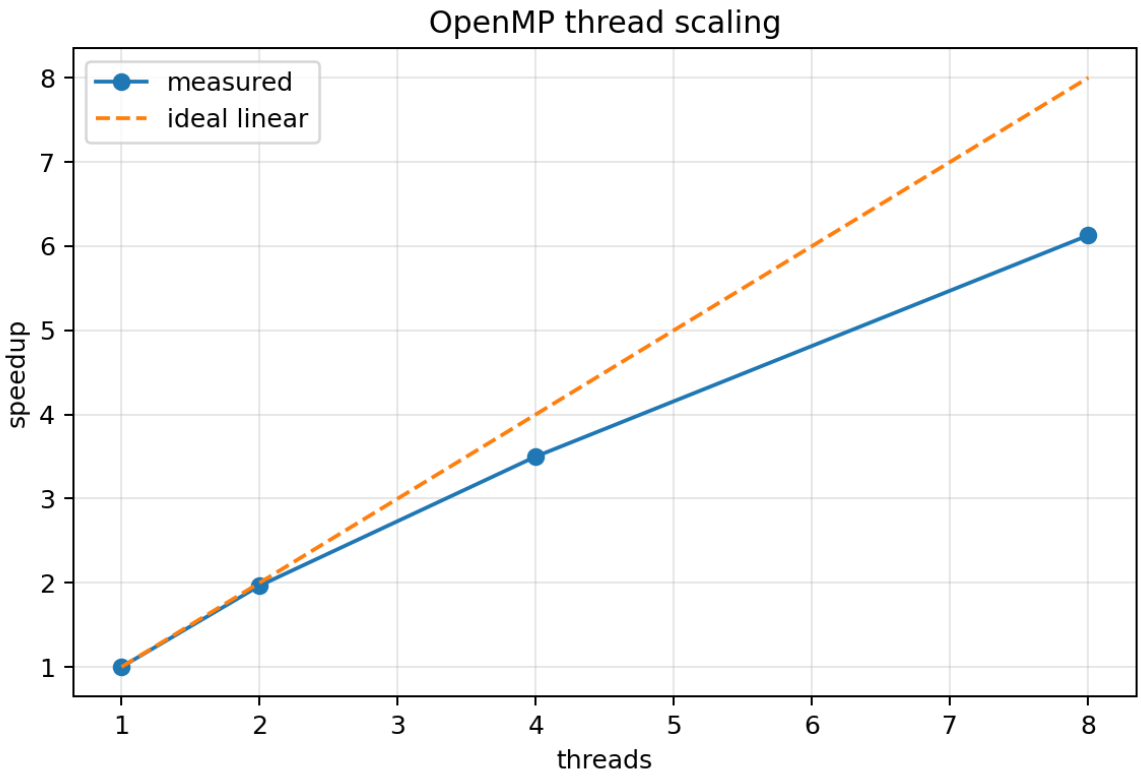
7.1 Historical Dirichlet Figures

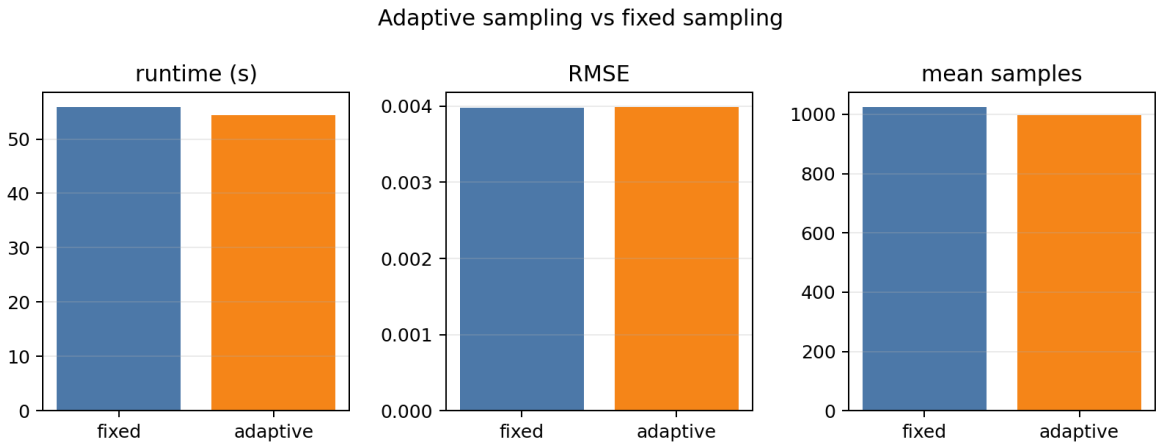
Zombie baseline	WoSt reference
	
	

7.2 Historical Neumann Figures

Zombie baseline	WoSt reference
	
	

7.3 Historical WoSt-Only Figures





8. Limitations and Careful Claims

Safe claims:

- The Bunny benchmark uses a high-resolution 70,580-face mesh.
- WoSt and Zombie closely agree on the clean Dirichlet analytic benchmark.
- Both methods show Monte Carlo convergence with increasing random walks.
- Mixed Neumann behavior is more implementation-sensitive than Dirichlet behavior.
- WoSt's Neumann implementation is faster than Zombie at practical epsilon values in the fresh formal run.
- WoSt now includes meaningful optimization diagnostics for adaptive sampling, antithetic sampling, lazy refinement, CRN, epsilon sensitivity, and normal convention.

Avoid overclaiming:

- Do not claim million-triangle scalability unless a million-triangle mesh is actually tested.
- Do not claim perfect linear speedup unless the thread sweep supports it for the target machine and workload.
- Do not treat the Zombie FCPW vs WoSt tiny_bvh timing as a pure backend-only comparison, because Zombie timing is measured through Python-loop calls.
- Do not claim that relative-standard-error adaptive sampling is fully tuned for Bunny-scale production until a full-size tuned run is completed.

9. Final Summary

The final combined evidence supports the following project story:

```
WoSt is accurate on the clean Dirichlet Bunny benchmark and agrees closely with
the Zombie baseline. The method also supports mixed Neumann boundaries, validated
first with a sphere/cube normal-convention sanity test and then benchmarked on
Bunny. Compared with Zombie, WoSt is slightly more accurate but slower in the
Dirichlet solve-loop formal run; in the mixed Neumann formal run, WoSt is faster
and competitive or better at practical epsilon values because it uses much
shorter reflected paths. Additional WoSt optimization experiments show that
relative-standard-error adaptive sampling can reduce sample counts, antithetic
sampling can reduce variance, and lazy star-radius refinement can dramatically
reduce geometric query cost without changing RMSE in the tested diagnostic
configuration.
```

10. Key Files

Reports:

```
C:\THU\homework\zombie\results_zombie_final_report\ZOMBIE_VS_WOST_FINAL_REPORT.m
d
experiments/optimization_report.md
experiments/formal_comparison_report.md
experiments/final_integrated_report.md
```

Formal CSVs:

```
experiments/formal_wost_dirichlet/results/benchmark_summary.csv
experiments/formal_wost_dirichlet/results/geometry_benchmark.csv
experiments/formal_wost_neumann/results/benchmark_summary.csv
experiments/formal_zombie_dirichlet/benchmark_summary.csv
experiments/formal_zombie_dirichlet/zombie_vs_wost_summary.csv
experiments/formal_zombie_neumann/benchmark_summary.csv
experiments/formal_zombie_neumann/zombie_vs_wost_neumann_summary.csv
```

Optimization figures:

```
experiments/figures/adaptive_rmse_errorbars.png
experiments/figures/adaptive_samples_errorbars.png
experiments/figures/antithetic_pointwise_variance.png
experiments/figures/antithetic_rmse_errorbars.png
experiments/figures/antithetic_variance_errorbars.png
experiments/figures/lazy_refinement_ratio_errorbars.png
experiments/figures/lazy_tradeoff_rmse_vs_refinement.png
experiments/figures/epsilon_sensitivity_histogram.png
experiments/figures/neumann_normal_diagnostics.png
```

Integrated local figure folder:

```
experiments/final_report_figures/
```